ABDALLAH ABOU HAJAL

AI-Drug Discovery | Cheminformatics | Pharmaceutical Sciences (MSc) | Data Analytics & AI Automation

Abu Dhabi, United Arab Emirates

Phone #: +971 56 884 9457 | **Email:** Abdallah.Abouhajal@gmail.com | **Date of Birth:** 18 June 1998 | **Marital Status:** Single

Personal Website: Linkedin Profile: Google Scholar: Scopus Profile:

Research Output: 15 publications | 133 citations | G-scholar h-index: 8

SUMMARY

Motivated researcher with an MSc in Pharmaceutical Sciences, specializing in AI for drug discovery and cheminformatics. Since completing my degree, I have been working in academic publishing, first as a Managing Editor, and currently as a Data Analyst/AI Automation lead, where I build AI-powered automation tools to streamline editorial workflows. With first-author publications in high-impact journals (JCIM, Expert Opinion on Drug Discovery), I am now seeking a PhD to further apply machine learning and data-driven methods to drug discovery challenges.

PROFESSIONAL EXPERIENCE

Data Analyst & AI Automation Lead, Scifiniti Publishing **May 2025 – Present**

- Building LLM-based AI agents to automate editorial workflows
- Developing Python-based tools to streamline publishing operations
- Creating automated dashboards for journal performance metrics
- Analyzing submission trends to identify bottlenecks and support strategic planning

Managing Editor, Scifiniti Publishing **Feb 2023 – May 2025**

- Managed day-to-day operations across 5+ STEM journals
- Served as primary contact for authors, reviewers, and editors to resolve conflicts and maintain review timelines
- Contributed to improving peer review workflows and author experience

MSc Student & Research / Lab Assistant, College of Pharmacy, Al Ain University **Sep 2021 - Feb 2023**

- Awarded a postgraduate scholarship (lab and research assistant duties).
- Contributed to research in drug discovery, cheminformatics, and machine learning applications.
- Delivered laboratory sessions and designed in-class activities for undergraduate students.
- Assisted in writing scientific manuscripts

EDUCATION

Master of Science in Pharmaceutical Sciences **Sep 2021 – Apr 2023**

College of Pharmacy, Al Ain University – Abu Dhabi, UAE

- Graduated with Excellent (GPA: 3.94)
- Thesis: **Use of machine learning in the development of a drug discovery tool for predicting promiscuous aggregate-based inhibitors**
  - Supervised by Prof. Mohammad Ghattas (Reader in Medicinal Chemistry)
  - Co-supervised by Prof. Boulbaba Ben Amor (Reader in Artificial Intelligence & Computer Science)
- Aggregation is an important source of false positives and wasted effort in early-stage drug discovery. The project employed (1) machine learning and ensemble modeling to predict small colloidally aggregating molecules from chemical structure, (2) large-scale data integration and feature selection to improve model performance and applicability, and (3) interpretability techniques to identify key molecular features driving aggregation. The final tool was deployed via a public web server for broad use in early-stage drug discovery.

Bachelor of Science in Pharmacy **Sep 2016 – Jun 2021**

College of Pharmacy, Al Ain University – Abu Dhabi, UAE

- Graduated with Excellent (GPA: 3.83)
- Honor Student – Academic Year 2017/2018

Science Major – Secondary School Certificate **Sep 2013 – Jun 2016**

Khalifa Bin Zayed High School – Abu Dhabi, UAE

- Graduated with 96.8% – Honor Student

AWARDS

First Place, BindHack Hackathon | Insilico Medicine, Masdar City **2025**

Built the best-performing model for antibody-antigen binding prediction in a 6-hour AI/drug discovery competition using smart feature engineering under tight time pressure.

Three Minute Thesis (Second Place), 2nd AAU Health and Biomedical Postgraduate Symposium | Al Ain University **2024**

Condensed my MSc thesis into a 3-minute talk for a general audience.

Best Quality Poster Award (First Place), 1st International Conference on Pharmacy and Biomedical Sciences | Al Ain University **2023**

Presented my research on ML-based aggregation prediction as a scientific poster.

TECHNICAL & RESEARCH SKILLS

Programming & Data Analysis

- Python (proficient)
- Classical ML: Pandas, NumPy, Matplotlib, scikit-learn, TensorFlow AutoGluon, CatBoost, LightGBM.
- LLM & Agentic AI: LLM API integration (Ollama, Anthropic, OpenAI), Microsoft Agent Framework, agentic workflow design.
- SPSS (basic proficiency)

Cheminformatics & Drug Discovery

- RDKit, MolVS, SHAP, programmatic MD simulation setup and trajectory analysis
- Machine learning & Chemical Language Models for Early-Stage Drug Discovery

Research & Professional Skills

- Full research lifecycle: hypothesis design → modeling → manuscript preparation
- Data-driven problem solving and strategic planning
- Project lead, workflow automation, and attention to detail.
- Cross-disciplinary communication and team collaboration

RESEARCH INTERESTS

- AI in drug discovery | Cheminformatics | Agentic AI & LLM applications in drug discovery | Predictive modeling | Explainable AI (XAI) | Chemical language models (CLMs) | Software development

LANGUAGES

- **English** (IELTS Academic 7.5 — L8.0, R8.5, W6.5, S7.5) | **Arabic** (Native)

PUBLICATIONS

1. **Abou Hajal A*.** Chemical language models for early-stage drug discovery: applications, pitfalls, and future directions. Journal of Computer-Aided Molecular Design. 2026;40:239. ***Corresponding author**
2. **Abou Hajal A*,** Al Meslamani AZ, Computational approaches to adverse outcome pathway networks: advances in construction, quantification, and regulatory decision support. Computational Toxicology .2026,100429 ***Corresponding author**
3. **Abou Hajal A*,** Bustanji L, Bryce RA, Ghattas MA. An introduction to GitHub and its significance for AI-Driven drug discovery. *Expert Opinion on Drug Discovery*.2026;2641511. ***Corresponding author**
4. **Abou Hajal A***, Sakkal M. Introducing AutoML framework for drug–drug interaction prediction: Application of AutoGluon. Toxicology Mechanisms and Methods. 2026;1–11. ***Corresponding author**
5. Sakkal M, **Abou Hajal A***. Large language models in drug delivery: A review of the current landscape and future perspectives. J Pharm Sci. 2025;115(2):104147. ***Corresponding author**
6. Sakkal M, **Abou Hajal A***. Machine learning predictions of tumor progression: How reliable are we? Computers in Biology and Medicine. 2025;191:110156. ***Corresponding author**
7. Al Meslamani AZ, **Abou Hajal A**. Language models for drug–drug interactions: current applications, pitfalls, and future directions. Expert Opinion on Drug Metabolism and Toxicology. 2025;21(9):1083–1102.
8. **Abou Hajal A**, Bryce RA, Amor BB, Atatreh N, Ghattas MA. Boosting the accuracy and chemical space coverage of the detection of small colloidal aggregating molecules using the BAD Molecule Filter. Journal of Chemical Information and Modeling. 2024;64(13):4991–5005.
9. **Abou Hajal A**, Al Meslamani AZ. Insights into artificial intelligence utilisation in drug discovery. Journal of Medical Economics. 2024;27(1):304–308.
10. **Abou Hajal A**, Al Meslamani AZ. Overcoming barriers to machine learning applications in toxicity prediction. Expert Opinion on Drug Metabolism and Toxicology. 2024;20(7):549–553.
11. Elnour AA, **Abou Hajal A**, Goaddar R, Elsharkawy N, Mousa S, et al. Exploring the pharmacy students' perspectives on off-campus online learning experiences amid COVID-19 crises: A cross-sectional survey. Saudi Pharmaceutical Journal. 2023;31(7):1339–1350.
12. Kharaba Z, AlAhmad MM, Elnour AA, **Abou Hajal A**, Abumweis S, et al. Are we ready yet for digital transformation? Virtual versus on-campus OSCE as assessment tools in pharmacy education. A randomized controlled head-to-head comparative assessment. Saudi Pharmaceutical Journal. 2023;31(3):359–369.
13. Al Mazrouei N, Elnour AA, **Abou Hajal A**, Sadeq A, Khalid AK, et al. A cross-sectional survey on the experience of people with allergic rhinitis amidst COVID-19 pandemic: The impact of the facemask: A pilot study. Pharmacy Practice. 2023;21(3):18.
14. Elnour AA, Raja NS, **Abou Hajal A**, et al. Protocol for systematic review and meta-analysis of randomized controlled trials, cost-benefit analysis and interrupted time-series interventions on pharmacist's prescribing. Pharmacy Practice. 2022;20(3):2713.

PUBLICATIONS

15. Hudaib MM, **Abu Hajal AF**, Sakkal MM. Chemical composition of the volatile oil from aerial parts of Rosmarinus officinalis L. growing in UAE. Journal of Essential Oil Bearing Plants. 2022;25(2):282–289

MANUSCRIPTS UNDER REVISION & ONGOING PROJECTS

1. Computational Flagging of Assay Interference in Drug Discovery: A Review of Rule-Based, Machine Learning, Graph Neural Networks, and Chemical Language Models **(Submitted to Journal of Computational Chemistry, Wiley)**
2. Aggregation Cliffs in Screening Benchmarks: A Matched Molecular Pair Census and a Pair-Level Evaluation of Published Filters **(Ongoing project)**
3. MAD: An Automated Molecular Dynamics Pipeline for Detecting Small Colloidally Aggregating Molecules **(Ongoing project)**